R&D | RetailPulse

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Project Charter and Business Problem

> **In simple words**
>
> This document explains why we do this project. We find where demand is hard to predict, so the business can make better stock decisions.

1. Business context

RetailPulse runs grocery stores in many cities. Every day, each store must decide how much of each product family to stock. A **forecast** is a guess of future sales. When the guess is too high, stock is wasted or sits unsold. When it is too low, shelves are empty and customers leave.

2. Problem statement

> **The decision problem**
>
> Supply Chain and Category teams need to know **where demand is hard to predict**, so they can decide where to add safety stock, where to review orders, and how to plan for holidays and promotions. Today they have no clear view of where forecasts miss, by how much, and in which direction.

> **Important assumption**
>
> The Kaggle data has no company forecast. We build a **simple baseline forecast** in SQL (same weekday last week). So our results show **where a simple baseline fails**, which means where demand is hard to predict. They are not the errors of an existing company forecast.

3. Business questions we must answer

#	Question	Type
A	Which product families are hardest to predict?	Where (category)
B	Which states, cities and store types are hardest?	Where (region)
C	When does error rise: months, holidays, paydays, promotions?	When (season)
D	Do we guess too high (excess stock) or too low (empty shelves)?	Direction
E	What should the business do about it?	Action

4. Decisions this project supports

Decision	Who decides	Evidence we give
Which families need a better forecast method first	Supply Chain Manager	Top high-error families and Pareto
Where to add safety stock	Category Manager	Under-forecast (shortfall) units by family and month
Where to cut orders to reduce excess	Supply Chain Manager	Over-forecast (excess) units by family and region
How to plan for holidays and promotions	Planning Team	Error on holiday, payday and promotion days
Which regions need support	Regional Manager	WAPE and Bias by state, city, store type

5. Scope

Included	Not included
Historical sales analysis (Kaggle train, stores, holidays, oil, transactions)	Machine learning or advanced forecasting models
Simple SQL baseline forecasts and error measurement	Changing the company buying process
Hypotheses, insights and recommendations	Money impact (no price or cost data)
Power BI dashboard, report, presentation	Real-time data pipelines

6. Success criteria

- At least 6 insights, each with evidence, a comparison, a reason, and a recommended action.
- Top high-error families named and ranked by share of total error.
- A dashboard where a manager finds the main message in 30 seconds.
- A one-page executive summary and a final report.
- All dashboard totals match the SQL totals.

7. Limits we state openly

- No price, cost or stock data, so impact is measured in **units**.
- The baseline is simple on purpose. Better methods may do better.
- Reasons for error are **likely causes** unless the data proves them.

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